

Course Syllabus

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 Edit

Some recommended sources:

[Martin Fowler](https://martinfowler.com/)  [\(https://martinfowler.com/\)](https://martinfowler.com/)

[Joel Spolsky](https://www.joelonsoftware.com/)  [\(https://www.joelonsoftware.com/\)](https://www.joelonsoftware.com/)

Course Summary:

Date	Details	Due
Tue Sep 5, 2023	 Flamingo Book chapter 1	due by 10am
	(https://courseworks2.columbia.edu/courses/179301/assignments/10802)	
Tue Sep 5, 2023	 4156 first day of classes	10:10am to 11:25am
	(https://courseworks2.columbia.edu/calendar?event_id=491945&include_contexts=course_179301)	
Thu Sep 7, 2023	 4156 class services are not apps	10:10am to 11:25am
	(https://courseworks2.columbia.edu/calendar?event_id=491960&include_contexts=course_179301)	
Tue Sep 12, 2023	 Coping with Hitchhikers and Couch Potatoes on Teams	due by 10am
	(https://courseworks2.columbia.edu/courses/179301/assignments/10801)	
Tue Sep 12, 2023	 4156 class working in teams	10:10am to 11:25am
	(https://courseworks2.columbia.edu/calendar?event_id=491946&include_contexts=course_179301)	
Thu Sep 14, 2023	 4156 class finding bugs	10:10am to 11:25am
	(https://courseworks2.columbia.edu/calendar?event_id=491961&include_contexts=course_179301)	

Date	Details	Due
Fri Sep 15, 2023	 end of program change period https://courseworks2.columbia.edu/calendar?event_id=492016&include_contexts=course_179301	12am
Mon Sep 18, 2023	 Assignment T0: Team Formation https://courseworks2.columbia.edu/courses/179301/assignments/10801	due by 11:59pm
Tue Sep 19, 2023	 4156 class RESTful APIs https://courseworks2.columbia.edu/calendar?event_id=491947&include_contexts=course_179301	10:10am to 11:25am
Thu Sep 21, 2023	 4156 class RPC-based APIs https://courseworks2.columbia.edu/calendar?event_id=491962&include_contexts=course_179301	10:10am to 11:25am
Mon Sep 25, 2023	 Assignment T1: Preliminary Project Proposal https://courseworks2.columbia.edu/courses/179301/assignments/10801	due by 11:59pm
Tue Sep 26, 2023	 4156 class finding bugs via execution https://courseworks2.columbia.edu/calendar?event_id=491948&include_contexts=course_179301	10:10am to 11:25am
Thu Sep 28, 2023	 4156 class more on APIs https://courseworks2.columbia.edu/calendar?event_id=491963&include_contexts=course_179301	10:10am to 11:25am
Tue Oct 3, 2023	 4156 class finish equivalence partitions https://courseworks2.columbia.edu/calendar?event_id=491949&include_contexts=course_179301	10:10am to 11:25am
Thu Oct 5, 2023	 4156 class unit testing https://courseworks2.columbia.edu/calendar?event_id=491964&include_contexts=course_179301	10:10am to 11:25am

Date	Details	Due
	 Assignment T2: Revised Project Proposal https://courseworks2.columbia.edu/courses/179301/assignments/10801	due by 11:59pm
Tue Oct 10, 2023	 4156 class mocking https://courseworks2.columbia.edu/calendar?event_id=491950&include_contexts=course_179301	10:10am to 11:25am
	 The Butterfly Effect https://courseworks2.columbia.edu/courses/179301/assignments/11372	due by 10am
Thu Oct 12, 2023	 4156 class integration testing https://courseworks2.columbia.edu/calendar?event_id=491965&include_contexts=course_179301	10:10am to 11:25am
Tue Oct 17, 2023	 4156 class logging https://courseworks2.columbia.edu/calendar?event_id=491951&include_contexts=course_179301	10:10am to 11:25am
	 middle of semester https://courseworks2.columbia.edu/calendar?event_id=492017&include_contexts=course_179301	12am
Thu Oct 19, 2023	 4156 class continuous integration https://courseworks2.columbia.edu/calendar?event_id=491966&include_contexts=course_179301	10:10am to 11:25am
	 Assignment T3: First Iteration https://courseworks2.columbia.edu/courses/179301/assignments/10801	due by 11:59pm
Tue Oct 24, 2023	 4156 class requirements https://courseworks2.columbia.edu/calendar?event_id=491952&include_contexts=course_179301	10:10am to 11:25am
Thu Oct 26, 2023	 4156 class CRC design https://courseworks2.columbia.edu/calendar?event_id=491967&include_contexts=course_179301	10:10am to 11:25am

Date	Details	Due
	 Assignment T4: First Iteration Demo https://courseworks2.columbia.edu/courses/179301/assignments/10801	due by 11:59pm
Fri Oct 27, 2023	 first individual assessment available https://courseworks2.columbia.edu/calendar?event_id=514416&include_contexts=course_179301	12am
Mon Oct 30, 2023	 First Individual Assessment https://courseworks2.columbia.edu/courses/179301/assignments/10802	due by 11:59pm
Tue Oct 31, 2023	 4156 class design principles https://courseworks2.columbia.edu/calendar?event_id=491953&include_contexts=course_179301	10:10am to 11:25am
Thu Nov 2, 2023	 4156 class refactoring https://courseworks2.columbia.edu/calendar?event_id=491968&include_contexts=course_179301	10:10am to 11:25am
Mon Nov 6, 2023	 university holiday https://courseworks2.columbia.edu/calendar?event_id=491980&include_contexts=course_179301	12am
Tue Nov 7, 2023	 university holiday no class https://courseworks2.columbia.edu/calendar?event_id=492073&include_contexts=course_179301	12am
Thu Nov 9, 2023	 4156 class design patterns https://courseworks2.columbia.edu/calendar?event_id=491969&include_contexts=course_179301	10:10am to 11:25am
Tue Nov 14, 2023	 4156 class continue design patterns https://courseworks2.columbia.edu/calendar?event_id=491955&include_contexts=course_179301	10:10am to 11:25am
Thu Nov 16, 2023	 Knightmare: A DevOps Cautionary Tale	due by 10am

Date	Details	Due
	https://courseworks2.columbia.edu/courses/179301/assignments/11538	
	 4156 class GCP deployment https://courseworks2.columbia.edu/calendar?event_id=491970&include_contexts=course_179301	10:10am to 11:25am
Tue Nov 21, 2023	 4156 class test order https://courseworks2.columbia.edu/calendar?event_id=491956&include_contexts=course_179301	10:10am to 11:25am
Wed Nov 22, 2023	 university holiday https://courseworks2.columbia.edu/calendar?event_id=491982&include_contexts=course_179301	12am
Thu Nov 23, 2023	 university holiday https://courseworks2.columbia.edu/calendar?event_id=492018&include_contexts=course_179301	12am
Fri Nov 24, 2023	 university holiday https://courseworks2.columbia.edu/calendar?event_id=492019&include_contexts=course_179301	12am
Tue Nov 28, 2023	 4156 class microservices and ROS https://courseworks2.columbia.edu/calendar?event_id=491957&include_contexts=course_179301	10:10am to 11:25am
Thu Nov 30, 2023	 4156 class UML basics https://courseworks2.columbia.edu/calendar?event_id=491972&include_contexts=course_179301	10:10am to 11:25am
	 Assignment T5: Second Iteration https://courseworks2.columbia.edu/courses/179301/assignments/10801	due by 11:59pm
Tue Dec 5, 2023	 4156 class test suite quality https://courseworks2.columbia.edu/calendar?event_id=491958&include_contexts=course_179301	10:10am to 11:25am

Date	Details	Due
Thu Dec 7, 2023	 4156 class last day of class (https://courseworks2.columbia.edu/calendar?event_id=491973&include_contexts=course_179301)	10:10am to 11:25am
	 Assignment T6: Second Iteration Demo (https://courseworks2.columbia.edu/courses/179301/assignments/10801)	due by 11:59pm
Fri Dec 8, 2023	 second individual assessment available (https://courseworks2.columbia.edu/calendar?event_id=514417&include_contexts=course_179301)	12am
Mon Dec 11, 2023	 last day of classes (https://courseworks2.columbia.edu/calendar?event_id=491985&include_contexts=course_179301)	12am
	 Second Individual Assessment (https://courseworks2.columbia.edu/courses/179301/assignments/10802)	due by 11:59pm
Tue Dec 12, 2023	 study days (https://courseworks2.columbia.edu/calendar?event_id=492052&include_contexts=course_179301)	12am
Wed Dec 13, 2023	 study days (https://courseworks2.columbia.edu/calendar?event_id=492009&include_contexts=course_179301)	12am
Thu Dec 14, 2023	 study days (https://courseworks2.columbia.edu/calendar?event_id=492010&include_contexts=course_179301)	12am
Fri Dec 15, 2023	 final exams (https://courseworks2.columbia.edu/calendar?event_id=491986&include_contexts=course_179301)	12am
Mon Dec 18, 2023	 final exams (https://courseworks2.columbia.edu/calendar?event_id=491989&include_contexts=course_179301)	12am

Date	Details	Due
	 Assignment T7: Demo Day https://courseworks2.columbia.edu/courses/179301/assignments/10801	due by 4pm
Tue Dec 19, 2023	 final exams https://courseworks2.columbia.edu/calendar?event_id=491990&include_contexts=course_179301	12am
Wed Dec 20, 2023	 final exams https://courseworks2.columbia.edu/calendar?event_id=491991&include_contexts=course_179301	12am
Thu Dec 21, 2023	 final exams https://courseworks2.columbia.edu/calendar?event_id=491992&include_contexts=course_179301	12am
Fri Dec 22, 2023	 final exams https://courseworks2.columbia.edu/calendar?event_id=491993&include_contexts=course_179301	12am
Mon Mar 11, 2024	 spring break https://courseworks2.columbia.edu/calendar?event_id=535337&include_contexts=course_179301	12am
Tue Mar 12, 2024	 spring break https://courseworks2.columbia.edu/calendar?event_id=535338&include_contexts=course_179301	12am
Wed Mar 13, 2024	 spring break https://courseworks2.columbia.edu/calendar?event_id=535339&include_contexts=course_179301	12am
Thu Mar 14, 2024	 spring break https://courseworks2.columbia.edu/calendar?event_id=535340&include_contexts=course_179301	12am
Fri Mar 15, 2024	 spring break https://courseworks2.columbia.edu/calendar?event_id=535341&include_contexts=course_179301	12am
	 4156 class https://courseworks2.columbia.edu/calendar?	

Date	Details	Due
	event_id=492058&include_contexts=course_179301)	
	 4156 class: APIs (https://courseworks2.columbia.edu/calendar?event_id=492072&include_contexts=course_179301) 	
	 4156 class: coding (https://courseworks2.columbia.edu/calendar?event_id=492045&include_contexts=course_179301) 	
	 4156 class: continuous integration (https://courseworks2.columbia.edu/calendar?event_id=492043&include_contexts=course_179301) 	
	 4156 class: CRC cards (https://courseworks2.columbia.edu/calendar?event_id=492067&include_contexts=course_179301) 	
	 4156 class: debugging (https://courseworks2.columbia.edu/calendar?event_id=492035&include_contexts=course_179301) 	
	 4156 class: design patterns (https://courseworks2.columbia.edu/calendar?event_id=492046&include_contexts=course_179301) 	
	 4156 class: design principles (https://courseworks2.columbia.edu/calendar?event_id=492037&include_contexts=course_179301) 	
	 4156 class: equivalence partitions (https://courseworks2.columbia.edu/calendar?event_id=492036&include_contexts=course_179301) 	
	 4156 class: finding bugs (https://courseworks2.columbia.edu/calendar?event_id=492044&include_contexts=course_179301) 	

 [4156 class: insecure coding](https://courseworks2.columbia.edu/calendar?event_id=492057&include_contexts=course_179301)
(https://courseworks2.columbia.edu/calendar?event_id=492057&include_contexts=course_179301)

 [4156 class: introduction](https://courseworks2.columbia.edu/calendar?event_id=492065&include_contexts=course_179301)
(https://courseworks2.columbia.edu/calendar?event_id=492065&include_contexts=course_179301)

 [4156 class: MVC](https://courseworks2.columbia.edu/calendar?event_id=492034&include_contexts=course_179301)
(https://courseworks2.columbia.edu/calendar?event_id=492034&include_contexts=course_179301)

 [4156 class: project issues](https://courseworks2.columbia.edu/calendar?event_id=492048&include_contexts=course_179301)
(https://courseworks2.columbia.edu/calendar?event_id=492048&include_contexts=course_179301)

 [4156 class: refactoring](https://courseworks2.columbia.edu/calendar?event_id=492027&include_contexts=course_179301)
(https://courseworks2.columbia.edu/calendar?event_id=492027&include_contexts=course_179301)

 [4156 class: running tests](https://courseworks2.columbia.edu/calendar?event_id=492038&include_contexts=course_179301)
(https://courseworks2.columbia.edu/calendar?event_id=492038&include_contexts=course_179301)

 [4156 class: SE4ML](https://courseworks2.columbia.edu/calendar?event_id=492064&include_contexts=course_179301)
(https://courseworks2.columbia.edu/calendar?event_id=492064&include_contexts=course_179301)

 [4156 class: test doubles](https://courseworks2.columbia.edu/calendar?event_id=492032&include_contexts=course_179301)
(https://courseworks2.columbia.edu/calendar?event_id=492032&include_contexts=course_179301)

 [4156 class: test oracles](https://courseworks2.columbia.edu/calendar?event_id=492042&include_contexts=course_179301)
(https://courseworks2.columbia.edu/calendar?event_id=492042&include_contexts=course_179301)

 [4156 class: testing overview](https://courseworks2.columbia.edu/calendar?event_id=492050&include_contexts=course_179301)
(https://courseworks2.columbia.edu/calendar?event_id=492050&include_contexts=course_179301)

**4156 class: UML**

(https://courseworks2.columbia.edu/calendar?event_id=492021&include_contexts=course_179301)

**4156 class: use cases and agile**

(https://courseworks2.columbia.edu/calendar?event_id=492060&include_contexts=course_179301)

**4156 class: user stories**

(https://courseworks2.columbia.edu/calendar?event_id=492039&include_contexts=course_179301)

**4156 class: working in teams**

(https://courseworks2.columbia.edu/calendar?event_id=492056&include_contexts=course_179301)

**academic holiday**

(https://courseworks2.columbia.edu/calendar?event_id=492003&include_contexts=course_179301)

**academic holiday**

(https://courseworks2.columbia.edu/calendar?event_id=492005&include_contexts=course_179301)

**Adjusted Roll Call**

(<https://courseworks2.columbia.edu/courses/179301/assignments/11774>)

**Adjusted Team Project**

(<https://courseworks2.columbia.edu/courses/179301/assignments/10801>)

**class cancelled**

(https://courseworks2.columbia.edu/calendar?event_id=492022&include_contexts=course_179301)

**CVN only**

(<https://courseworks2.columbia.edu/courses/179301/assignments/11261>)

**demo day**

(<https://courseworks2.columbia.edu/calendar?>)

Date	Details	Due
	event_id=492001&include_contexts=course_179301)	
	 Differential Factor (https://courseworks2.columbia.edu/courses/179301/assignments/10802)	
	 Edstem and/or Zoom Comments (https://courseworks2.columbia.edu/courses/179301/assignments/10802)	
	 end of program change period (https://courseworks2.columbia.edu/calendar?event_id=492000&include_contexts=course_179301)	
	 first day of classes (https://courseworks2.columbia.edu/calendar?event_id=492033&include_contexts=course_179301)	
	 First Individual Assessment Ends (https://courseworks2.columbia.edu/calendar?event_id=492006&include_contexts=course_179301)	
	 First Individual Assessment Starts (https://courseworks2.columbia.edu/calendar?event_id=492007&include_contexts=course_179301)	
	 last day of classes (https://courseworks2.columbia.edu/calendar?event_id=492029&include_contexts=course_179301)	
	 Roll Call Attendance (https://courseworks2.columbia.edu/courses/179301/assignments/10802)	
	 second individual assessment posted (https://courseworks2.columbia.edu/calendar?event_id=492063&include_contexts=course_179301)	
	 study day (https://courseworks2.columbia.edu/calendar?)	

Date	Details	Due
	event_id=492023&include_contexts=course_179301)	
	 study day https://courseworks2.columbia.edu/calendar?event_id=492055&include_contexts=course_179301)	
	 thanksgiving https://courseworks2.columbia.edu/calendar?event_id=492004&include_contexts=course_179301)	
	 This is a Fake Assignment https://courseworks2.columbia.edu/courses/179301/assignments/11538	
	 university holiday https://courseworks2.columbia.edu/calendar?event_id=492002&include_contexts=course_179301)	
	 university holiday https://courseworks2.columbia.edu/calendar?event_id=492066&include_contexts=course_179301)	